CSE 1st Year Syllabus

Computer Science & Engineering • Rajshahi University of Engineering & Technology

CSE 1100

Computer Fundamentals and Ethics Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Computer Science and Engineering as a Discipline: Computer Science, Computer Engineering, Theoretical Computer Science, Central Themes of CSE, Subfields of CSE, Course Map for Four Year Undergrad CSE Degree, Role of Mathematics in CSE, Role of Electrical Engineering in CSE, Diversified Applications of CSE: Artificial Intelligence, Machine Learning, Digital Image Processing, Computer Graphics, Computer Security etc.

Hardware: Types and Generation of Computers, Data Representation, Computer Arithmetic, Processor, Types of Memory, Peripherals, Interfacing, Assembler, Compiler, Interpreter, Levels of Programming Language, Data Communication and Computer Network.

Software: Computer Programming, Data Structure & Algorithms, Database System, Types of Software, Software Licenses, Software Engineering, Familiarization with Various Operating Systems (Windows, DOS, UNIX, Android, IOS Etc.), Computer Operations: Text Processing (MS-WORD, etc.), Spreadsheet (MS-EXCEL etc.), Browser Software (Chrome, Firefox etc.).

Ethics: Introduction to Ethics and Morality, Ethics for IT Workers and IT Users, Privacy, Intellectual Property, Ethical Decisions in Software Development, Ethics of IT Organizations, Cybercrimes, Cyberattacks and Cybersecurity, Laws against Cybercrimes, Professional Responsibility.

CSE 1101

Structured Programming

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Introduction to Structured Programming: Algorithm, Writing, Debugging and Running Programs using Compiler / Interpreter.

Basic Concepts: Basic I/O, Formatted I/O, Different Data Types and their Range, Operators and their Precedence, Operands and Expression, Expression Evaluation, Conditional Logic, Switch-Case, Character, ASCII Value, Reading and Writing Character, Integer to Character Conversion.

Loop: Basic of Loop, while Loop, for Loops, do-while Loop, Entry Controlled and Exit Controlled Loops, Nested Loop, Formulating Problems using Loops.

Array: Basics of Array, Necessity, Declaration, Initialization, Array Manipulation- Accessing through Indices, Accessing using Loops, One-Dimensional and Multi-Dimensional Arrays.

String: Basic String Operations, Difference between String and Character-Array, String I/O, Array of Strings.

Pointer: Introduction to Pointer, Pointer Operations, Pointers and Array, Array of Pointers.

Functions: Defining & Calling of User-defined Functions, Void Functions with No Parameters, Functions with Return Type and Parameters, Call by Value, Local and Global Variables, Scope of Variables, Built-in Functions, Recursive Functions, Passing Arrays in a Function as Parameter, Call by Reference.

Custom data types: Structures, Unions, Enumerations.

File: Basic Files Operations, Opening, Closing and Updating Binary and Sequential Files, File I/O (Read from File and Write in File).

Advanced Topics: Operations on Bits, Preprocessors and Macros.

CSE 1102

Structured Programming Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Structured Programming: Introduction, Basics Concepts, Selection Statements, Control Statements, Arrays, Strings, Pointer, Functions, Custom Data Types, Files, Advanced Topics.

EEE 1151

Basic Electrical Engineering

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Direct Current: Voltage, Current, Resistance and Power, Laws of Electrical Circuits, Methods of Network Analysis, Capacitance, Types of Capacitors, Capacitors in Series and Parallel, Inductance, Types of Inductors, Faraday's Law and Lenz's Law, Inductors in Series and Parallel.

Alternating current: Signal and Waveforms, Instantaneous and RMS Values of Current, Voltage and Power, Average Power, AC Analysis for Various Combination of R, L and C Circuits, Phasor Representation of Sinusoidal Quantities, Resonance, Frequency Response, Passive Filters.

EEE 1152

Basic Electrical Engineering Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Sessional on Basic Electrical Engineering: Basic of Electrical Circuit, Experiments on Circuit Theorems, Circuit Simulation and Design.

Math 1113

Differential and Integral Calculus

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Differential Calculus: Limit, Continuity and Differentiability, Differentiation of Explicit and Implicit Function and Parametric Equations, Significance of Derivatives, Differentials, Successive Differentiation of Various Types of Functions, Leibnitz's Theorem, Rolle's Theorem, Mean Value Theorem, Taylor's Theorem in Finite and Infinite Forms, Maclaurin's Theorem in Finite and Infinite Forms, Lagrange's Form of Remainders, Cauchy's Form of Remainder, Expansion of Functions by Differentiation and Integration. Partial Differentiation, Euler's Theorem, Tangent and Normal, Maxima and Minima, Points of Inflection and their Applications, Evaluation of Indeterminate Forms by L'Hospital Rule, Curvature, Evaluate and Inviolate, Asymptotes, Envelopes, Curve Tracing.

Integral Calculus: Definitions of Integration, Integration by the Method of Substitutions, Integration by the Method of Successive Reduction, Definite Integrals, Beta Function and Gamma Function, Area under a Plane Curve in Cartesian and Polar Coordinates, Area of the Region Enclosed by Two Curves in Cartesian and Polar Coordinates, Parametric and Pedal Equations, Intrinsic Equation, Volumes of Solids of Revolution, Volume of Hollow Solids of Revolution by Shell Method, Area of Surface of Revolution, Multiple Integration.

Hum 1113

Functional English

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Grammar: Construction and Transformation of Sentences, Analysis of Sentence Structure, Use of Preposition, Question Words, WH & Yes/No Question, Phrases & Idioms, Correction, Conditional Sentences, Punctuation, Pronunciation, Phonetic Transcription, Spoken English, Vocabulary.

Reading: Techniques and Strategies for Improving Comprehension Skills, Prose Pieces by Renowned Authors.

Writing: Paragraph and Essay Writing, Amplification, Précis Writing.

Commercial/Professional Correspondence: Business Letters, CV, Notices for and Minutes of Meetings, Tenders & Schedules, Memos & Press Release, Research Paper/Technical Report Writing.

Hum 1114

Functional English Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Sessional on Functional English: Principles of Language Communication, Academic Vocabulary for Scholarly Reading-Writing, Listening for Monologue and Dialogue, Note Taking and Drafting Skills, Common Speaking Functions, English Varieties: Pronunciation, Intonation, Stress, Reading Strategies for Non-literary Texts, Multi-Skill and integrated Questions, Produce and Design Audio and Video Presentation, Sample Project: Short biography, Sample Conference Presentation.

Chem 1113

Inorganic and Physical Chemistry

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Chemical Bond: Ionic Bond, Covalent Bond, Coordination Bond, Metallic Bond, Hydrogen Bond, Dipole Bond, Vander Waal's Forces, Hybridization, Resonance, Valence Bond Theory (VBT), Molecular Orbital Theory (MOT), Linear Combination of Atomic Orbital (LCAO) Method.

Thermo-chemistry: Types of Energy, Enthalpy of Reaction, Heat of Combustion, Heat of Formation and Heat of Neutralization, Experimental Determination of Thermal Changes during Chemical Reaction.

Solution: Types of Solution, Factors Influencing the Solubility of Substance, Mechanism of Dissolution, Solution of Gases in Liquids, Different Units of Concentration, Distribution Law and its Application, Properties of Dilute Solution, Raoult's Law - its Application, Elevation of Boiling Point, Depression of Freezing Point and Osmotic Pressure.

Conductivity of Electrolytic Solution: Type of Conductors, Conductance, Specific Conductance, Equivalent Conductance, Mechanism of Electrolytic Conductance, Factors Influencing Conductivity, Arrhenius Theory, Law of Independent Migration of Ions and its Applications, Determination of Transport Number, Abnormal Conductance.

Electromotive Forces: Electrochemical Cell, Cell Reaction, Cell Potential, Cell Representation, Measurement of EMF of a Cell, Relation Between EMF and Free Energy, Electrode Potential, Electrochemical Series, Nernest's Equation, Different Types of Reference Electrodes and pH Measurement, Over Potential, Lithium Ion Battery, Fuel Cell its Latest Development.

Photochemistry: Photochemical Reactions, Laws of Photochemistry, Quantum Yield and its Determination, Photosensitized Reaction, Photo-Physical Processes.

Spectroscopy: Quantization of Energy, Basic Elements of Spectroscopy.

Chemistry of Polymerization: Classification, Bonding in Polymer, Thermosetting and Thermoplastic Polymer, Synthesis, Properties and Uses of Some Polymers-Polyethylene, PVC, Bakelite, and Melamine etc.

Chem 1114

Inorganic and Physical Chemistry Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Sessional on Inorganic and Physical Chemistry: Introduction to Lab Experiments, Conducting Lab Experiments, Preparing Lab Report and Presentation.
